# Chapter 4 System Analysis and Design

## 4.1 Introduction

This chapter presents the analysis and design of the Intelligent Agent Identity Verification System. It specifies the required registration, identity checking, decision, access control and administrative services; the quality, data and interface conditions supporting those services; and the models used to communicate the solution. It also presents interface wireframes and a requirements traceability and verification plan. Implementation evidence and observed evaluation results are reported in Chapter 5.

The software design uses Object-Oriented Analysis and Design (OOAD) to describe actors, domain objects and interactions. The classifier is supported by an experimental machine learning workflow that separates model development from evaluation. The selected views are a use case diagram, class diagram, entity relationship diagram, system architecture diagram, sequence diagram, database schema and interface wireframes. Together they address the human workflow, evidence processing, data relationships and presentation of decisions. The machine learning data design is documented separately because a transactional database diagram does not describe training and evaluation separation.

## 4.2 System Requirements

The requirements define required behaviour, quality conditions, data obligations and integration boundaries. The catalogue follows the requirements-engineering focus of ISO/IEC/IEEE 29148:2018. Quality categories are informed by ISO/IEC 25010:2023; these references guide organisation and evaluation and are not a claim of certification (ISO/IEC/IEEE, 2018; ISO/IEC, 2023).

Requirement identifiers are stable references. FR denotes a functional requirement, NFR a quality or evaluation requirement, DR a data requirement and IR an interface requirement. H, M and L indicate high, medium and low priority within the project. Source codes are A for agent task needs, B for administrator review needs, O for operational integrity and protection, and R for research reproducibility and evaluation. A stated verification method is a planned check, not a reported pass. Chapter 5 records the evidence actually obtained.

**Table 4.1: Stakeholder needs used to organise the requirements**

| Stakeholder or need | Design responsibility |
|---|---|
| Agent | Complete onboarding, understand failures and see the current outcome without accessing another agent's data. |
| Administrator | Inspect exception cases and decision evidence, make accountable interventions and retrieve audit history. |
| System maintainer | Protect credentials and stored evidence, handle failures and manage retention and recovery. |
| Research evaluation | Preserve data/model provenance, isolate evaluation and report consequential classification errors. |

### 4.2.1 Functional Requirements

The functional requirements describe the actor, required behaviour and expected verification. Registration establishes the account and personal profile; evidence capture then supplies the document and live selfie. Automated processing produces a decision, while administrators handle exceptions. A client-side capture challenge and model-based liveness analysis serve different parts of this workflow.

**Table 4.2: Functional requirements**

| ID<br>Priority<br>Source | Requirement | Verification plan |
|---|---|---|
| | **Agent registration and authentication** | |
| FR-01<br>H / A | The system shall authenticate an account through email and password or Google sign-in before allowing protected agent functions. | C4-T01: valid, invalid, expired and unconfirmed account scenarios. |
| FR-02<br>H / A | The system shall allow an authenticated agent to submit full name, phone number, national ID number and date of birth as a personal profile. | C4-T02: required fields, invalid dates and persisted profile values. |
| FR-03<br>H / A | The system shall accept identity evidence as JPEG or PNG images up to 5 MiB per file for the supported national ID, passport and driving licence capture flows. | C4-T03: supported formats, file-size boundary and document-side cases. |
| FR-04<br>H / A | The system shall guide the agent through live camera selfie capture and a displayed face movement or blink challenge before continuing the capture workflow. | C4-T04: successful challenge, camera refusal and failed capture. |
| FR-05<br>H / O | The system shall reject creation of an agent profile whose national ID number is already assigned to another profile. | C4-T02: duplicate identifier returns a conflict without a second profile. |
| FR-06<br>M / O | The system shall reject registration attempts using an email domain on the configured disposable-domain block list. | C4-T02: blocked and permitted domains; inspect rejection record. |
| | **Verification pipeline** | |
| FR-07<br>H / A | The system shall assess document and selfie blur, brightness and contrast before the full OCR, face matching and liveness pipeline proceeds. | C4-T05: clear and degraded images produce quality measurements. |
| FR-08<br>H / A | The system shall stop downstream verification when the image quality gate fails and identify the image and condition requiring recapture. | C4-T05: quality failure prevents subsequent model calls. |
| FR-09<br>H / A | The system shall compare OCR-extracted identity fields with registered details and return text-agreement evidence and a confidence score. | C4-T06: matched, mismatched and unreadable name, ID and date fields. |
| FR-10<br>H / A | The system shall compare the document portrait with the captured selfie and return a numerical face-match score. | C4-T06: matching and non-matching pairs, and a missing-face case. |

| ID<br>Priority<br>Source | Requirement | Verification plan |
|---|---|---|
| FR-11<br>H / A | The system shall assess presentation-attack evidence from the selfie and return a liveness score. | C4-T06: genuine and photo-presentation samples; record limitations. |
| FR-12<br>H / O | The system shall retain the six pipeline scores against the verification request as processing progresses. | C4-T07: inspect stored quality, OCR, face and liveness values. |
| FR-13<br>H / R | The system shall classify a complete valid six-score vector as Verified, Manual Review or Rejected using the available decision model. | C4-T08: validate input contract, class mapping and prediction output. |
| | **Access control and guardrails** | |
| FR-14<br>H / A | The system shall permit the verified-agent workspace only when the current authorised access decision grants access. | C4-T09: verified agent can enter the protected workspace. |
| FR-15<br>H / B | The system shall place Manual Review cases in the administrator queue while withholding verified-agent workspace access. | C4-T09: review case is visible to administrators and cannot enter workspace. |
| FR-16<br>H / O | The system shall withhold verified-agent workspace access when the current decision is Rejected. | C4-T09: rejected account cannot use the protected workspace. |
| FR-17<br>M / O | The system shall block further applicable registration attempts from an IP address when its recorded failures reach the configured limit within the configured window. | C4-T10: default limit of 3 failures in 24 hours and boundary cases. |
| FR-18<br>M / O | The system shall apply the configured failure limit to registration attempts associated with the same device fingerprint. | C4-T10: repeated fingerprint, expiry of window and separate devices. |
| | **Administrator management and audit** | |
| FR-19<br>H / B | The system shall allow an authorised administrator to inspect review cases with their available identity images and verification scores. | C4-T11: review queue, detail panel and expired-image state. |
| FR-20<br>H / B | The system shall record an authorised administrator approval or rejection and apply the corresponding access outcome. | C4-T11: decision, status, access record and audit entry agree. |
| FR-21<br>M / B | The system shall allow an administrator to retrieve an agent's verification history and associated audit trail. | C4-T11: distinguish multiple attempts and retain previous outcomes. |
| FR-22<br>L / B | The system shall export administrator-selected audit records as CSV using date-range and agent filters. | C4-T12: filtered rows, valid CSV and no unauthorised export. |
| FR-23<br>M / B | The system shall notify connected administrator interfaces when a case enters Manual Review. | C4-T12: new or updated review case and connection recovery. |
| FR-24<br>H / O | The system shall append an audit entry for each defined significant authentication, registration, verification, decision and administrator action. | C4-T13: compare expected action sequence with recorded events. |

| ID<br>Priority<br>Source | Requirement | Verification plan |
|---|---|---|
| FR-25<br>H / O | Each audit entry shall identify its action, outcome, time and responsible user or system, with request, agent and network context where available. | C4-T13: validate event values, including events before profile creation. |
| FR-26<br>H / O | The system shall prevent ordinary authenticated agent and administrator clients from updating or deleting audit entries. | C4-T13: negative client write tests and policy inspection. |
| | **Current interface and operational capabilities** | |
| FR-27<br>M / A | The system shall support account confirmation and password recovery without granting access solely because a recovery request was submitted. | C4-T01: confirmation, reset request, expired link and successful recovery. |
| FR-28<br>M / A | The system shall allow an authenticated user on a supported device to enrol a passkey. | C4-T14: enrolment, cancellation and unsupported-device behaviour. |
| FR-29<br>M / A | The system shall authenticate a registered passkey using a fresh server challenge and the credential public key. | C4-T14: valid assertion, invalid signature and expired or reused challenge. |
| FR-30<br>M / A | The system shall allow users to list, rename and remove their own passkeys under the account recovery checks. | C4-T14: ownership, rename, removal and available recovery method. |
| FR-31<br>H / A | The system shall allow permitted recapture of failed evidence without requiring resubmission of evidence that remains acceptable. | C4-T05: document-only and selfie-only recapture; retry limits. |
| FR-32<br>M / A | The system shall display an agent's own verification progress, decision and available action in the identity workspace. | C4-T15: processing, review, verified, rejected and error states. |
| FR-33<br>M / B | The administrator directory shall support searching and status filtering, including verified agents whose image audit window is open. | C4-T11: search, filter and three-business-day window boundary. |
| FR-34<br>M / B | The system shall display available model status and saved evaluation results to authorised administrators. | C4-T16: available report, loading failure and fallback state. |

The repeated-attempt defaults are three recorded failures per IP or device within 24 hours; they are configurable guardrail choices and must be tested at their boundaries. Evidence is limited to JPEG or PNG at 5 MiB per file, where 1 MiB equals 1,048,576 bytes. Completing account registration or a selfie challenge does not itself confer verified-agent access.

### 4.2.2 Non Functional Requirements

The quality requirements cover performance measurement, security, privacy, reliability, interaction quality, maintainability and model evaluation. Performance is specified through a reproducible measurement protocol. An operational response-time acceptance limit must be justified using measurements in the intended environment and stakeholder needs before it is treated as a service commitment. Cold-start and steady-state behaviour must be reported separately.

**Table 4.3: Quality and evaluation requirements**

| ID<br>Priority<br>Source | Requirement | Verification plan |
|---|---|---|
| | **Performance and readiness** | |
| NFR-01<br>H / A | The evaluation shall measure complete pipeline elapsed time over ten sequential single-user runs in a declared environment, separating warm and cold starts. | C4-T17: record start/end events, median, maximum and all failures. |
| NFR-02<br>M / A | The evaluation shall measure representative non-pipeline API response times under a declared workload and record the host and network conditions. | C4-T17: authentication, profile, status and administrator queries. |
| NFR-03<br>M / O | Readiness checks shall distinguish process availability from availability of the model components needed for verification. | C4-T18: service running with model absent, loading or ready. |
| | **Security and privacy** | |
| NFR-04<br>H / O | Protected agent operations shall reject missing or invalid authentication and prevent cross-agent access. | C4-T01 and C4-T09: absent/expired token and other-agent identifiers. |
| NFR-05<br>H / B | Administrator operations shall reject an authenticated user without the administrator role. | C4-T09: agent token against every administrator route family. |
| NFR-06<br>H / A | Identity images shall remain private; an authorised signed URL shall last at most one hour and shall not outlive an applicable verified-case audit window. | C4-T19: public access denied; expiry and remaining-window boundaries. |
| NFR-07<br>H / O | Production network connections carrying credentials or identity evidence shall use HTTPS. | C4-T19: inspect frontend, backend, AI and storage endpoints. |
| NFR-08<br>H / O | Database client access shall be restricted by ownership and role; privileged service credentials shall remain on the server. | C4-T19: policy and configuration inspection plus cross-agent tests. |
| | **Reliability and failure handling** | |
| NFR-09<br>H / O | Threshold fallback shall be permitted only when valid complete scores exist and the SVM decision service is unavailable or returns an unusable prediction. | C4-T08 and C4-T18: fallback has a recorded cause and decision basis. |
| NFR-10<br>H / O | An incomplete or failed evidence-processing step shall not be represented as a successful completed verification. | C4-T18: inject quality, OCR, face, liveness and storage failures. |

| ID<br>Priority<br>Source | Requirement | Verification plan |
|---|---|---|
| | **Interaction quality and human oversight** | |
| NFR-11<br>M / A | The interface shall label each onboarding step and identify the current step, completed steps and the next available action. | C4-T15: walkthrough; record task completion and assistance needed. |
| NFR-12<br>H / A | Validation and quality failures shall identify the affected field or image and provide a corrective action. | C4-T05 and C4-T15: invalid fields, blur, denied camera and service error. |
| NFR-13<br>H / B | Each decision shall retain its basis so administrators can distinguish a model result, threshold fallback, quality failure and manual intervention. | C4-T07 and C4-T08: persist and retrieve the correct provenance. |
| NFR-14<br>H / B | Manual overrides shall require administrator authority and retain the actor, time and resulting access outcome. | C4-T11 and C4-T13: approve/reject override and trace resulting records. |
| NFR-15<br>M / O | Image quality, OCR, face matching, liveness and decision services shall expose separate contracts that can be checked independently. | C4-T20: module inspection and representative contract tests. |
| | **Model evaluation and accessibility** | |
| NFR-16<br>H / R | Model evaluation shall report per-class precision, recall and F1, macro F1, accuracy and a confusion matrix on an isolated evaluation set. | C4-T21: metrics derived from saved predictions and declared labels. |
| NFR-17<br>H / R | Model evaluation shall report Rejected-to-Verified and Manual-Review-to-Verified errors separately from aggregate accuracy. | C4-T21: explicit numerators, denominators and error interpretation. |
| NFR-18<br>H / R | Evaluation shall state the synthetic population assumptions and the untested demographic, camera, document and attack conditions. | C4-T21: limitations and provenance inspection; no unsupported fairness claim. |
| NFR-19<br>M / A | Essential form, navigation and review controls shall expose labels, keyboard operation, visible focus and understandable status text. | C4-T15: keyboard and screen-reader checks, including modal focus. |
| NFR-20<br>M / A | The interface shall preserve essential tasks on desktop and mobile layouts without losing field labels, feedback or access restrictions. | C4-T15: representative desktop and phone widths; both display themes. |

Model quality is assessed through the cost-relevant errors as well as aggregate metrics. A comparative improvement on synthetic data is not an absolute operational approval threshold. Deployment acceptance for real applicants requires justified error tolerances and representative evidence. The interface review uses selected WCAG 2.2 concerns, including labels, keyboard use, focus and status feedback, without claiming full conformance from wireframe inspection alone (World Wide Web Consortium, 2023).

### 4.2.3 Data Requirements

The transactional data comprise account references, agent profiles, document-object references, verification attempts, six-score records, decisions, access outcomes and audit events. The authentication provider manages passwords and sessions. Application password fields must not be presented as a second store of plaintext credentials. Optional passkeys store credential identifiers, public keys and supporting authentication metadata; the application does not receive the device's local fingerprint or face template through WebAuthn.

Table 4.4 specifies the data obligations. A requirement describes the intended design and its verification; it does not assert that the corresponding operational control has already passed. In particular, informed collection, retention beyond verified cases and backup restoration require explicit policy and evaluation evidence.

**Table 4.4: Data requirements**

| ID<br>Priority<br>Source | Requirement | Verification plan |
|---|---|---|
| DR-01<br>H / A | Profile data shall link a unique user account to personal identity details, validating required values and dates before persistence. | C4-T02: missing fields, duplicate identifier and invalid/future dates. |
| DR-02<br>H / A | Each stored evidence object shall be associated with an agent and an evidence type, including national ID front/back where required and selfie. | C4-T03: allowed image types, references and correct object ownership. |
| DR-03<br>H / R | The six-score feature vector shall use the names, order, units and permitted ranges defined in Table 4.5. | C4-T08: missing, non-numeric, non-finite and out-of-range inputs. |
| DR-04<br>H / O | Requests, scores, decisions and access records shall preserve stable keys and valid relationships while allowing an incomplete attempt to remain incomplete. | C4-T07: uniqueness, foreign keys and interrupted processing. |
| DR-05<br>H / B | Audit records shall retain structured action context with nullable agent/request links where no corresponding profile or request exists. | C4-T13: registration rejection, authentication and pipeline event cases. |
| DR-06<br>H / O | Verified-agent evidence images shall expire three business days after the latest successful verification, including manual approval. | C4-T22: Monday-Friday, Africa/Nairobi, weekend and repeated-approval cases. |
| DR-07<br>H / O | Expired verified-case image cleanup shall preserve document metadata, scores, decisions and audit records and retry failed object removals. | C4-T22: failed removal, cleared references and surviving metadata. |
| DR-08<br>H / O | Retention periods for pending, review and rejected evidence and retained metadata shall be explicitly defined before operational rollout. | C4-T22: policy review; these cases are excluded from current automatic purge. |
| DR-09<br>H / A | The onboarding process shall communicate the purpose, access conditions and retention of identity evidence before collection. | C4-T15: inspect disclosure and informed acknowledgement; evidence still required. |

| ID<br>Priority<br>Source | Requirement | Verification plan |
|---|---|---|
| DR-10<br>H / R | Synthetic data generation shall retain its configuration, seed, source-measurement provenance, label assumptions and duplicate-screening results. | C4-T23: regenerate from configuration and inspect recorded metadata. |
| DR-11<br>H / R | Training, calibration, selection and evaluation partitions shall be separated so evaluation records do not fit transformations or select candidates. | C4-T23: split manifest, training folds and frozen selection artifacts. |
| DR-12<br>M / O | The maintainer shall retain controlled backups of necessary metadata, configuration and model artifacts and verify a restoration procedure. | C4-T24: isolated restore and integrity check; record retention and exclusions. |

### Verification feature contract

The classifier consumes the ordered features in Table 4.5. Raw image-quality values are not interchangeable with arbitrary zero-to-one ratings. The fitted model pipeline performs its defined transformations. When the backend aggregates document and selfie quality, the resulting score must retain the same physical definition and unit used in development. Missing, non-finite or invalid values must not be silently converted to successful evidence.

**Table 4.5: Six-feature input contract**

| Feature | Meaning | Range | Interpretation |
|---|---|---|---|
| face_match_score | Face similarity | 0 to 100 | Document portrait versus selfie; score, not a probability of identity. |
| liveness_score | Presentation evidence | 0 to 1 | Selfie anti-spoof estimate; evaluation is limited to tested attacks. |
| ocr_confidence_score | Text agreement | 0 to 1 | Agreement between extracted and registered fields; not proof of authenticity. |
| blur_score | Laplacian variance | Finite and non-negative | Raw sharpness measure; larger values generally indicate sharper images. |
| brightness_score | Mean HSV value | 0 to 255 | Raw image brightness. |
| contrast_score | Grayscale standard deviation | 0 to 127.5 | Raw image contrast. |

### Training and evaluation data design

The decision-model development dataset contains 10,000 synthetic records: 6,000 Verified, 2,000 Manual Review and 2,000 Rejected. These are experimental class quotas rather than estimates of real applicant prevalence. Each record consists of six observed scores and one assumed adjudication label. The earlier 4,000-record dataset is retained as historical work and is not the final experiment described here.

The generator uses stored local measurements from Labeled Faces in the Wild for face matching, NUAA for real/photo presentation scores, and MIDV document frames and OCR measurements. The measurement files and their provenance inform score distributions. They do not provide joint three-class KYC ground truth. The synthetic labels, relationships between channels and uncertain-case scenarios are modelling assumptions. Historical measurement reuse also does not establish that source images were reprocessed with the current pipeline.

The original stratified split is 7,000 training, 1,500 validation and 1,500 test records. For version 3 development, the validation partition is divided into 750 calibration and 750 selection records. The original test partition is retired because earlier results informed the follow-up research questions. A fresh 1,500-record evaluation sample, comprising 900 Verified, 300 Manual Review and 300 Rejected cases, is drawn after candidate selection is frozen. This creates 11,500 unique experimental records across the original and fresh datasets, not an 11,500-record training set.

Preprocessing is fitted within training folds during cross-validation. Calibration fits only on its designated partition, and candidate/policy selection uses the separate selection partition. A stored configuration, seed lineage, source hashes, split manifests and duplicate checks support reproduction. The fresh evaluation sample is generated from the same simulator assumptions; it is not independent real-world validation. Its use for a predeclared release check must be distinguished from tuning on evaluation results.

Class probabilities must remain distinguishable from threshold-rule outputs. A threshold decision may use 0/1 indicator values for compatibility, but these are not calibrated certainty estimates. The model remains static during a verification request; completed cases do not automatically retrain it. Any later retraining requires a separate offline process using appropriately governed and confirmed data.

**Retention and recovery**

Verified-case images are retained for three business days after the latest successful decision, including manual approval. Business days are Monday to Friday in Africa/Nairobi; public holidays are not excluded. For example, a Friday approval at 14:00 expires on Wednesday at 14:00. Cleanup is checked at startup and every 15 minutes, so scheduled deletion depends on a running backend. Failed removals are retried; signed links must not extend access beyond the permitted window.

Cleanup removes the image objects and clears their references while retaining metadata, scores, decisions and audit history. Pending, review and rejected cases are excluded from this automatic policy. Their retention periods and the lifetime of retained metadata remain policy decisions requiring explicit definition. A recovery procedure must also account for storage copies and backups so restored data do not silently reintroduce expired evidence. No paid backup capability or completed restore test is assumed from the use of a hosted database.

### 4.2.4 Interface and Integration Requirements

The design separates the Vue presentation layer, Express application layer, Flask AI layer and Supabase persistence and identity services. The hosted arrangement uses Vercel for the frontend, Render for the backend and Microsoft Azure Container Apps for the AI service. Supabase provides authentication, PostgreSQL, object storage and database-change events. This deployment allocation preserves the logical separation; hosting success, resource settings and measured performance belong in Chapter 5.

The browser communicates with the application API using JSON or multipart uploads and bearer authentication. Email/password and Google authentication use Supabase Auth. The backend verifies identity, role and object ownership before protected operations. Server-held Supabase and AI credentials are not included in browser code. Camera permission and browser support are prerequisites for the live capture flow; password and Google recovery paths remain available when passkeys are unsupported.

The backend coordinates separate quality, OCR, face, liveness and SVM requests. Quality, OCR, face and liveness endpoints consume image data; the SVM endpoint consumes the six-score JSON vector. For the hosted AI service, a server-held API key accompanies requests. The configured SVM call timeout defaults to five seconds. A uniform bounded timeout and retry policy for every other integration is a design verification item and is not assumed to exist merely because calls use exception handling.

An unavailable SVM can trigger a logged threshold decision only after a complete valid score vector exists. Failure of the whole AI service before evidence extraction cannot be repaired by classifying missing scores. Such a failure returns a controlled error or incomplete state. Automatic retries of writes must not create duplicate profiles, requests, decisions or access records; safe retry behaviour must be checked at each operation boundary.

Files are written to private storage and retrieved as buffers for processing. Authorised administrator image views use signed URLs lasting no more than one hour and, for verified cases, no longer than the remaining image audit window. Socket.io conveys review notifications after database INSERT or UPDATE events place a request into review. Recipient authorisation, reconnect behaviour and consistency with the persisted queue are separate verification concerns.

**Table 4.6: Interface and integration requirements**

| ID<br>Priority<br>Source | Requirement | Verification plan |
|---|---|---|
| IR-01<br>H / A | The browser interface shall support labelled onboarding, role-appropriate navigation and inline validation on desktop and mobile devices. | C4-T15: wireframe walkthrough and implementation accessibility checks. |
| IR-02<br>H / O | Protected backend requests shall carry a valid Supabase bearer token and be checked for the required role and resource ownership. | C4-T01 and C4-T09: 401/403 and cross-agent scenarios. |

| ID<br>Priority<br>Source | Requirement | Verification plan |
|---|---|---|
| IR-03<br>H / O | Backend-to-AI calls shall use the defined REST contracts, with multipart images or JSON scores and structured JSON responses. | C4-T20: each endpoint input/output contract and invalid response. |
| IR-04<br>H / O | Calls to the hosted AI service shall authenticate using the server-held API credential over HTTPS. | C4-T19: absent/wrong credential, secret location and transport inspection. |
| IR-05<br>H / O | The decision-service call shall use its configured timeout; other integration calls shall have an explicitly documented failure and retry policy. | C4-T18: timeout, service refusal and bounded retry behaviour. |
| IR-06<br>H / O | The storage integration shall upload and retrieve private objects through authorised server operations and issue bounded signed image links. | C4-T03 and C4-T19: upload failure, private retrieval and expiry. |
| IR-07<br>M / B | Review notifications shall connect database review-state events to the administrator interface and recover from a disconnected client. | C4-T12: insert/update transition, toast and reconnect; check recipient access. |
| IR-08<br>M / A | Passkey integration shall use browser WebAuthn support and server verification while leaving established recovery sign-in methods available. | C4-T14: browser capability, challenge verification and recovery. |
| IR-09<br>H / A | Camera capture shall request permission, report an unavailable camera and show the preview used to produce the submitted evidence. | C4-T04 and C4-T15: allow/deny permissions and capture framing. |
| IR-10<br>M / B | The administrator evaluation interface shall label saved metrics with their model and dataset context and show an explicit unavailable-report state. | C4-T16: report metadata, missing report and active/fallback indicator. |

## 4.3 System Analysis Diagrams

### 4.3.1 Use Case Diagram

The use case view identifies the agent and administrator services within the identity-verification workflow. It relates registration and evidence submission to FR-01 to FR-13, controlled access to FR-14 to FR-16, and administration and auditing to FR-19 to FR-26. Its purpose is to clarify responsibility and the service boundary.

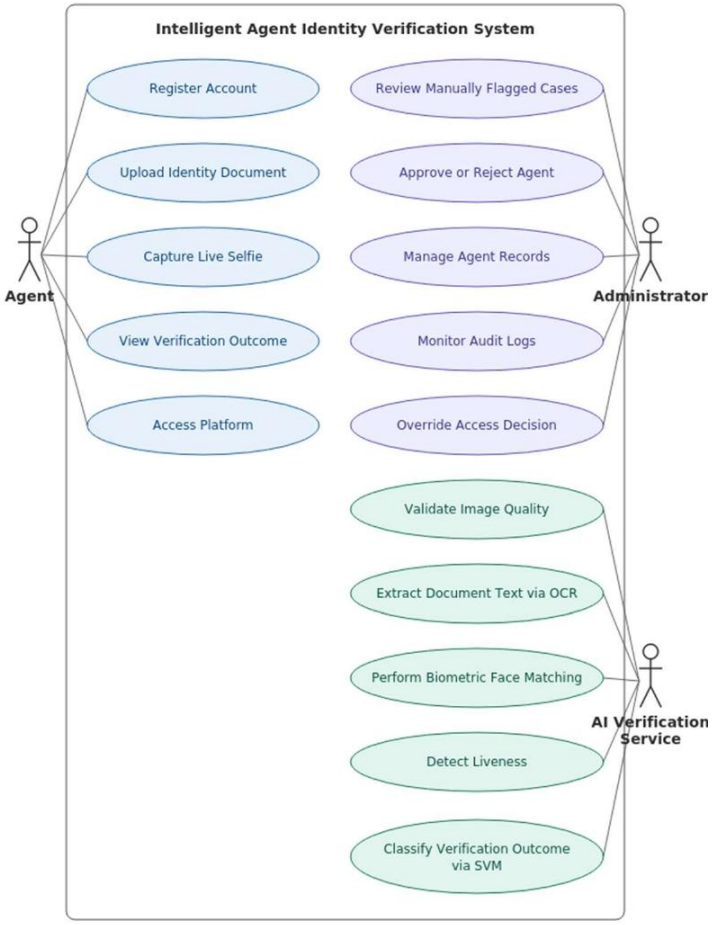



**Figure 4.1: Use Case Diagram**

The agent initiates onboarding and checks the outcome; the administrator reviews exceptions and inspects the audit history. Automated evidence processing supports those user goals. Eligibility for the protected workspace is a business precondition evaluated from the access decision. It must not be inferred solely from the fact that an outcome screen has been visited. The AI processing layer is a system component in the architecture described in Section 4.4.3.

## 4.4 System Design Diagrams

### 4.4.1 Class Diagram

The class view organises the domain concepts used to describe accounts, agents, documents, verification attempts, processing results, decisions and access. It supports the separation of responsibilities required by FR-07 to FR-13 and NFR-15. The conceptual model provides vocabulary for implementation without requiring every concept to become a separate database table.

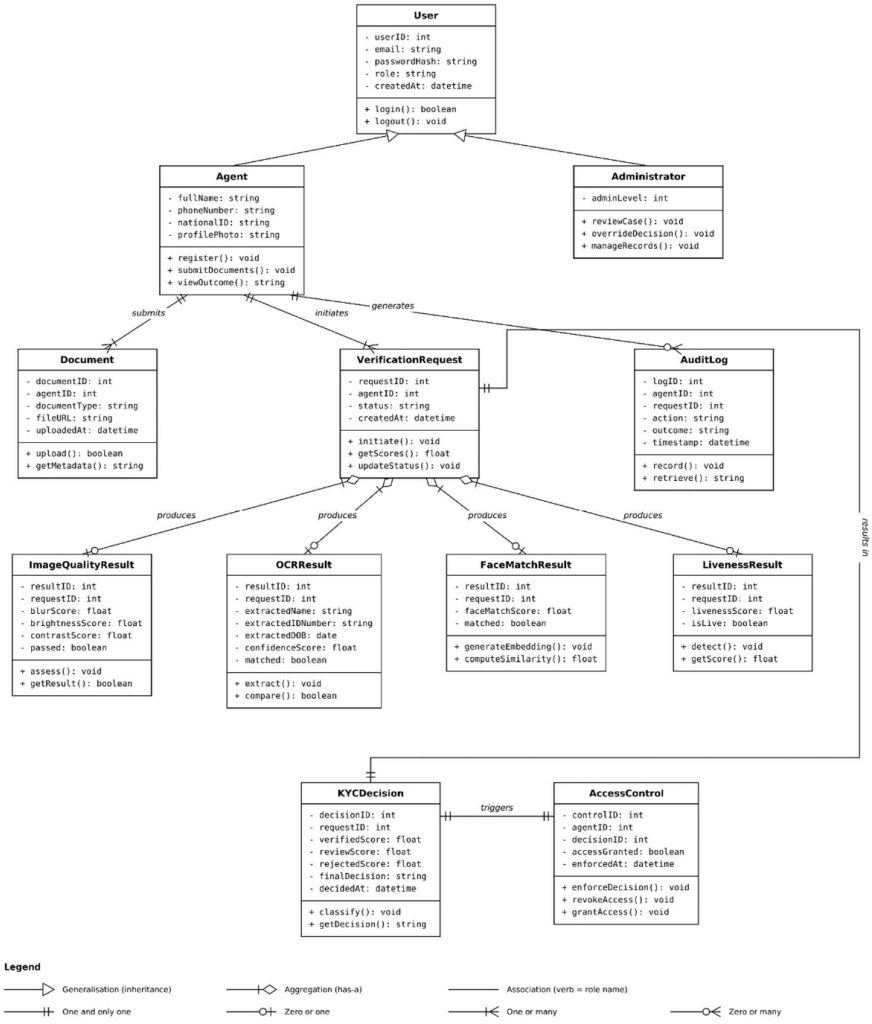



**Figure 4.2: Class Diagram**

A verification attempt provides the context for image-quality, OCR, face-match and liveness results. The decision consumes completed evidence, while access enforcement consumes the decision. Audit events refer to the relevant agent or request when those objects exist. Conceptual ownership must be implemented through explicit keys, validation and transaction boundaries; a diagram relationship alone does not prevent orphaned or partially written records.

### 4.4.2 Entity Relationship Diagram

The entity relationship view describes the persistent registration and verification records and their associations. It supports DR-01 to DR-05 by linking identity evidence and decisions to the relevant agent and request. The logical entity catalogue in Section 4.4.5 identifies the attributes and integrity rules needed by the current design.

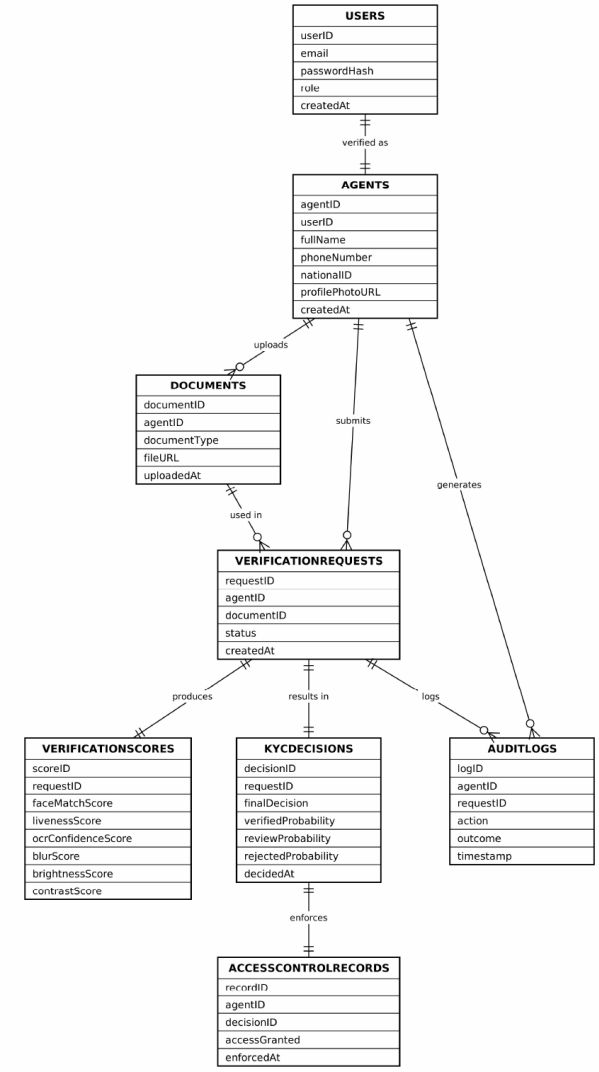



**Figure 4.3: Entity Relationship Diagram**

An account may exist before an agent profile is completed, and a request may exist before scores or a decision have been produced. A unique request reference permits at most one associated score row or decision row; it does not force the child row to exist. This distinction permits honest representation of pending and failed attempts. Audit references may be absent for authentication or blocked registration events that occur before profile creation.

### 4.4.3 System Architecture Diagram

The architecture view separates presentation, application coordination, evidence inference and persistent records. The four logical layers support the interface requirements in Table 4.6 and allow the verification services to be evaluated independently. The hosted service allocation does not change the responsibilities of these layers.

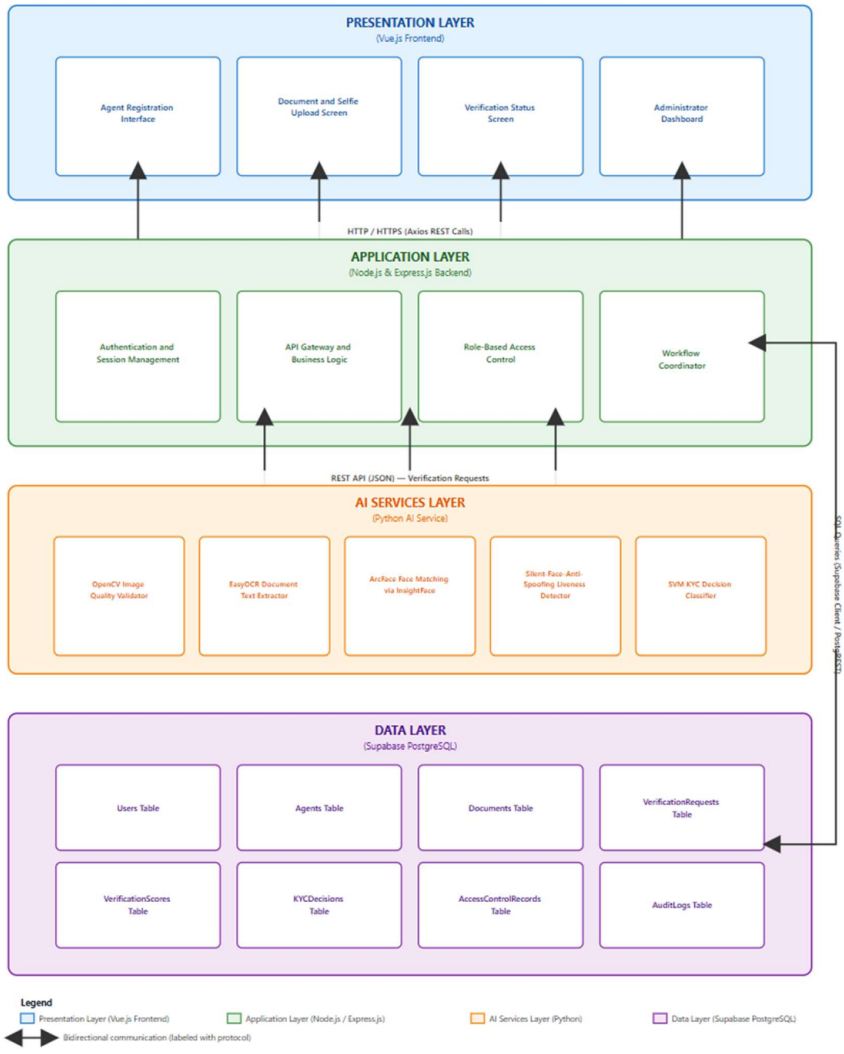



**Figure 4.4: System Architecture Diagram**

The frontend sends authenticated requests to the backend, which coordinates AI calls and persistence. Responses return through the relevant interfaces. Supabase authentication and database-change events introduce additional interactions, while review notifications travel from the backend to the administrator interface. Model-stage fallback depends on valid existing scores; a complete evidence-service outage instead leaves an error or incomplete attempt. Layer separation allows replacement behind a stable contract, rather than guaranteeing that any component change requires no integration work.

### 4.4.4 Sequence Diagram

The sequence view describes the progression from evidence submission to a recorded outcome and administrator review where required. It supports FR-07 to FR-16 and FR-24. The runtime design uses the backend to coordinate the individual AI endpoints and to associate their results with a verification request.

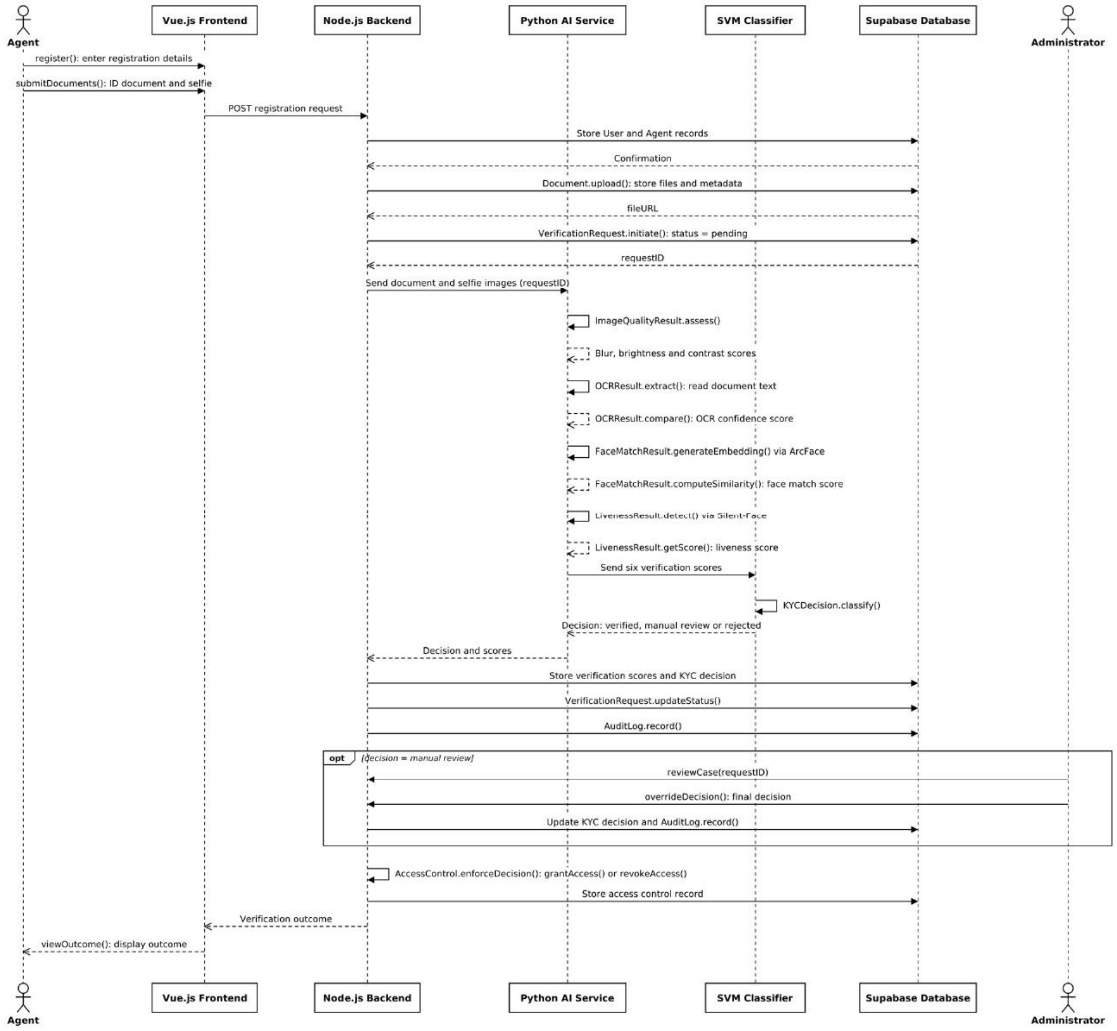



**Figure 4.5: Sequence Diagram**

Quality assessment gates subsequent processing. The backend stores the quality record and updates it as OCR, face and liveness evidence becomes available, then requests classification and applies the access decision. Audit events accompany the defined significant steps. A failed early step does not become a successful classifier input, and a manual-review outcome withholds verified workspace access. Administrator intervention is a separate authorised action that changes the recorded decision and access state.

### 4.4.5 Database Schema

The schema view presents the persistent entities, while Table 4.7 provides the logical keys, attributes and integrity conditions required by the current system. The logical design is the focus of this chapter; exact SQL declarations, indexes and deployed policy scripts are implementation evidence for Chapter 5.

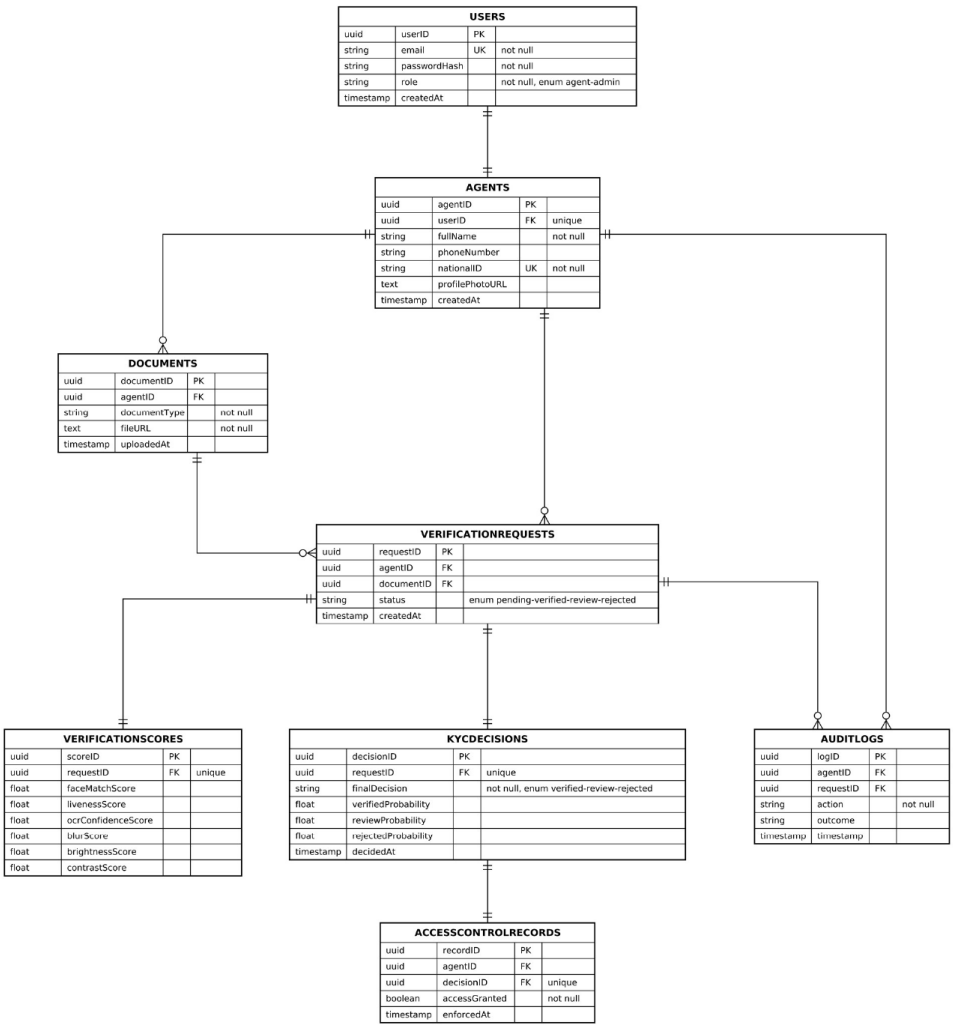



**Figure 4.6: Database Schema**

Profile, evidence, verification and decision records are separated to avoid repeating personal data in every attempt. Unique keys prevent duplicate associations, foreign keys maintain valid references, and allowed status values constrain state representation. Access-control policy and audit append-only behaviour require separate enforcement. Restricting inserts does not by itself prohibit updates or deletion. Privileged service operations remain outside ordinary-client row policies, so administrator database privileges must be governed separately.

Table 4.7 records the logical schema independently of implementation-specific SQL types. Each identifier denotes a stable key. The operational schema uses UUID identifiers; the logical rules allow an uncompleted account or attempt to exist without its later records.

**Table 4.7: Logical entity and integrity catalogue**

| Entity | Important attributes | Relationship and integrity rule |
|---|---|---|
| users | user_id; email; role; created_at | Account reference; unique email; identity is linked to Supabase Auth. |
| agents | agent_id; user_id; full_name; phone_number; national_id; date_of_birth | user_id is unique; an account has zero or one agent profile. national_id is unique. |
| documents | document_id; agent_id; document_type; file_url; uploaded_at | An agent may have several evidence records. Expired images can have cleared file references. |
| verification requests | request_id; agent_id; document_id; status; created_at | Each request belongs to an agent and references its identity evidence. |
| verification scores | score_id; request_id; six score attributes | request_id is unique. An attempt may have no score record before processing. |
| kyc decisions | decision_id; request_id; final_decision; class probabilities; decision_basis; decided_at | request_id is unique. A decision is absent until processing or intervention produces one. |
| access control records | record_id; agent_id; decision_id; access_granted; enforced_at | decision_id is unique; the record reflects the enforced outcome. |
| audit logs | log_id; agent_id; request_id; action; outcome; details; performed_by; ip_address; timestamp | Agent/request references may be absent for early events. Ordinary clients cannot alter entries. |
| passkeys and authentication support | Credential public key and identifier; user reference; counter; label; device metadata; challenge/rate-limit records | Optional authentication extension. Private keys and device biometrics remain outside the application. |

The core separation reduces duplication: account identity, personal profile, evidence metadata and repeated verification attempts are distinct. Derived scores and decisions reference the attempt instead of copying the profile. Structured audit details support varied event context without adding a column for every event type. The optional passkey support entities extend authentication rather than the classifier feature set. Exact deployed policies and schema migrations require implementation-level inspection.

### 4.4.6 User Interface Design and Wireframes

The wireframes show the principal screens and states using grayscale controls and bracketed data placeholders. They contain no applicant records and document the current design without claiming that they preceded implementation or establish usability. Each view links to the requirements it supports.

Agents move from authentication through details, document capture, selfie and outcome. Administrators navigate the directory, review, audit and model views. The desktop sidebar collapses into a mobile menu; paired fields stack in reading order. Only an authorised verified outcome permits workspace access.

#### 4.4.6.1 Account registration and sign in

The authentication wireframe separates account creation from sign-in. It provides the entry point for an agent or administrator before protected functions are offered. Registration gathers the account details, while sign-in presents password, Google and supported passkey options.

*Requirements addressed: FR-01, FR-06, FR-27 to FR-29; NFR-04; IR-02 and IR-08.*

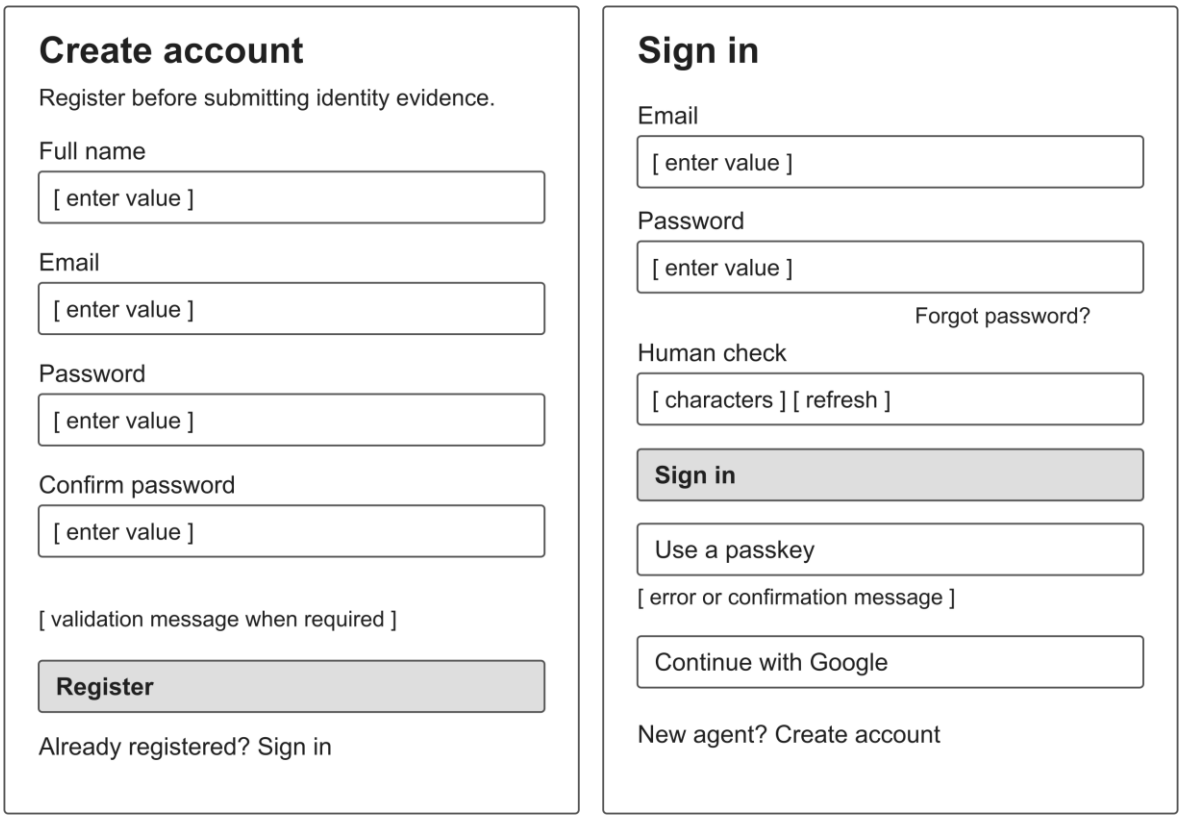



**Figure 4.7: Account registration and sign in wireframe**

Persistent labels identify the required inputs and a dedicated message region exposes validation or authentication failures. Password recovery and confirmation messages retain a route back to sign-in. The human-check control reflects the current password-login interface; its accessible alternative remains part of accessibility evaluation. Passkey cancellation returns control to established sign-in methods. The two panels represent separate screens, not one combined form.

#### 4.4.6.2 Personal profile details

The profile wireframe is the first protected onboarding step. It links the authenticated email to the identity details later compared with the document and displays the four-stage progress indicator.

*Requirements addressed: FR-02 and FR-05; NFR-11, NFR-12; DR-01; IR-01.*

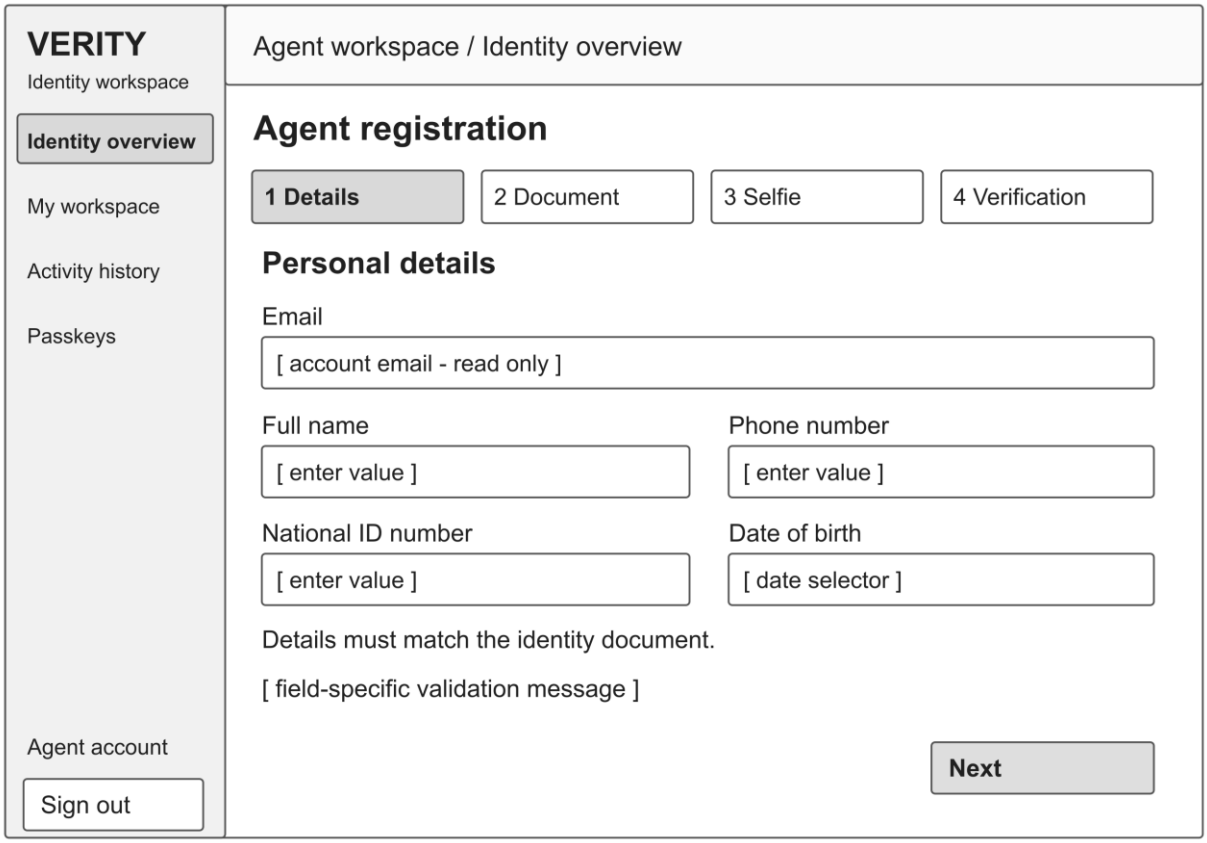



**Figure 4.8: Personal profile details wireframe**

The account email is read-only. Full name, phone number, national ID and date of birth remain grouped as one task, with validation beside the affected input. The Next action persists valid details before advancing. On a narrow screen, the paired fields stack in reading order. Existing values are restored when the user returns, and a duplicate national ID produces an explicit conflict rather than a second profile.

#### 4.4.6.3 Identity document capture

The document wireframe combines document-type selection, camera or file input, preview and image-quality feedback. It represents the selected capture step; national ID front and back are captured in sequence, while passport and driving-licence flows use their relevant document face.

*Requirements addressed: FR-03, FR-07, FR-08 and FR-31; DR-02; IR-06 and IR-09.*

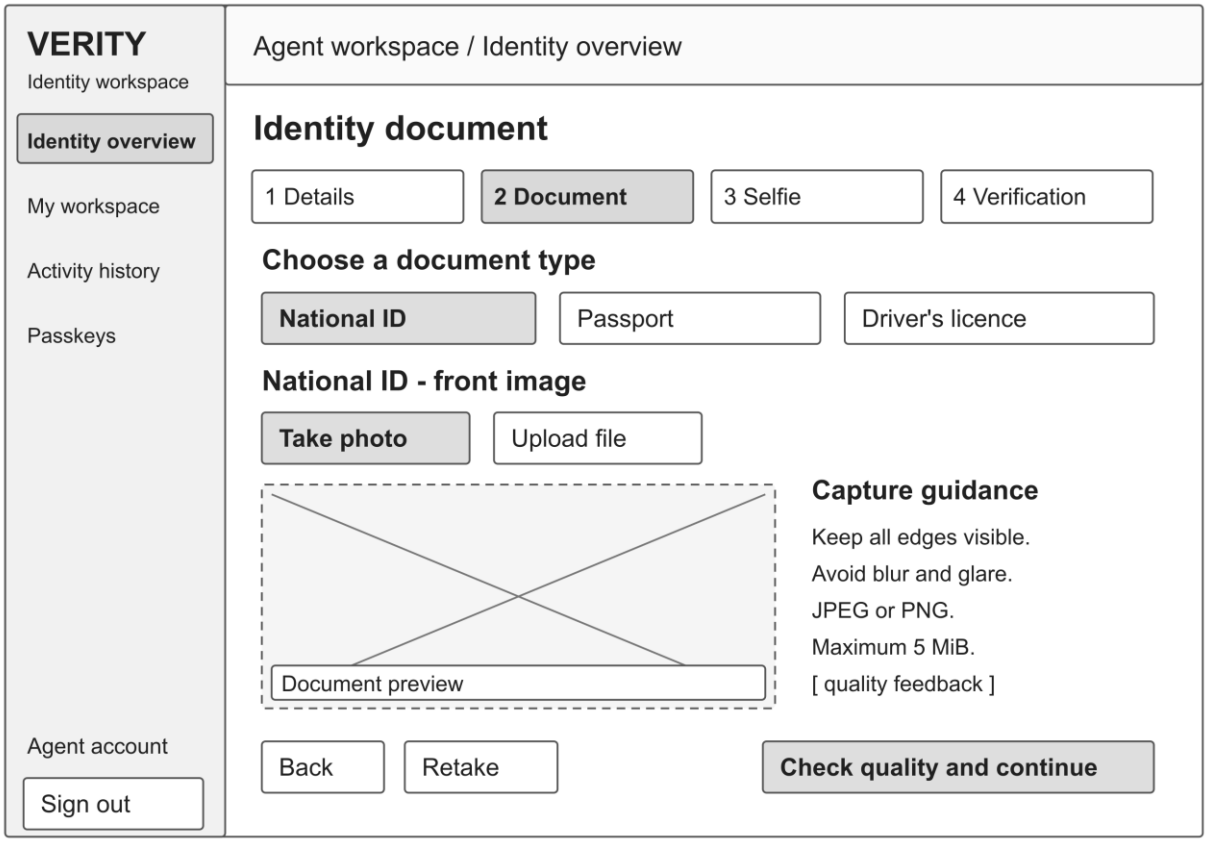



**Figure 4.9: Identity document capture wireframe**

Capture guidance is placed beside the preview and quality feedback remains visible before continuation. A user may retake or replace an image. The action changes from quality checking to continuation when the check succeeds. Rejected type/size, permission denial and service errors keep the user on the affected step. A successful image-quality check improves suitability for processing but does not establish that the document is authentic.

#### 4.4.6.4 Live selfie and challenge

The selfie wireframe provides a live preview, a face-position guide and the instruction for the current capture challenge. It separates capture readiness from later model-based liveness scoring.

*Requirements addressed: FR-04 and FR-11; NFR-11 and NFR-12; IR-09.*

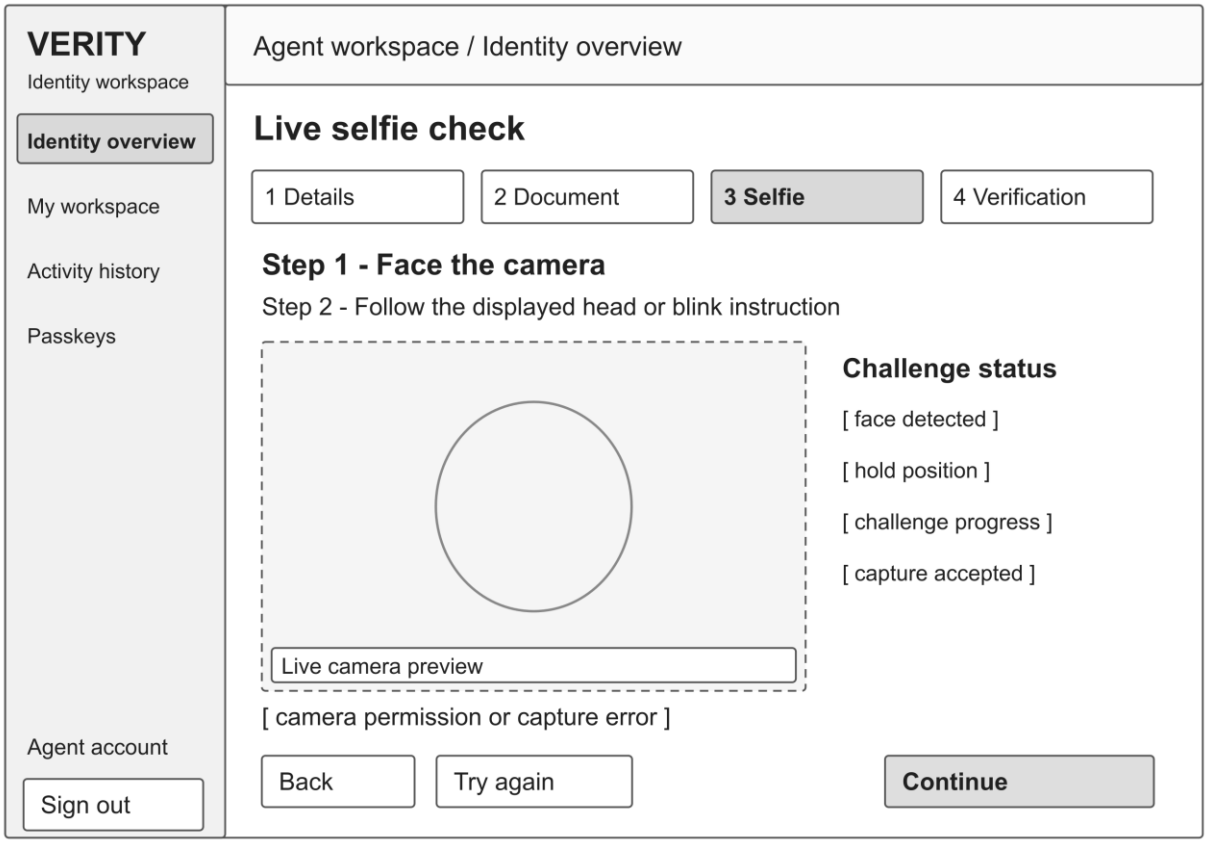



**Figure 4.10: Live selfie and challenge wireframe**

The first instruction asks the user to face the camera; the next displays the selected movement or blink task. Status text reports progress without relying only on colour. The user can retry after a failed capture or return to the previous step. A camera refusal must explain how to restore permission. Completing a client capture challenge is not a guarantee against every presentation attack and must not be reported as such.

#### 4.4.6.5 Verification progress and decision states

The verification wireframe presents the processing state and the principal outcome alternatives. The four state panels are shown together for design inspection; the operational interface displays the state relevant to the current attempt.

*Requirements addressed: FR-08 and FR-13 to FR-16, FR-31 and FR-32; NFR-09 to NFR-13.*

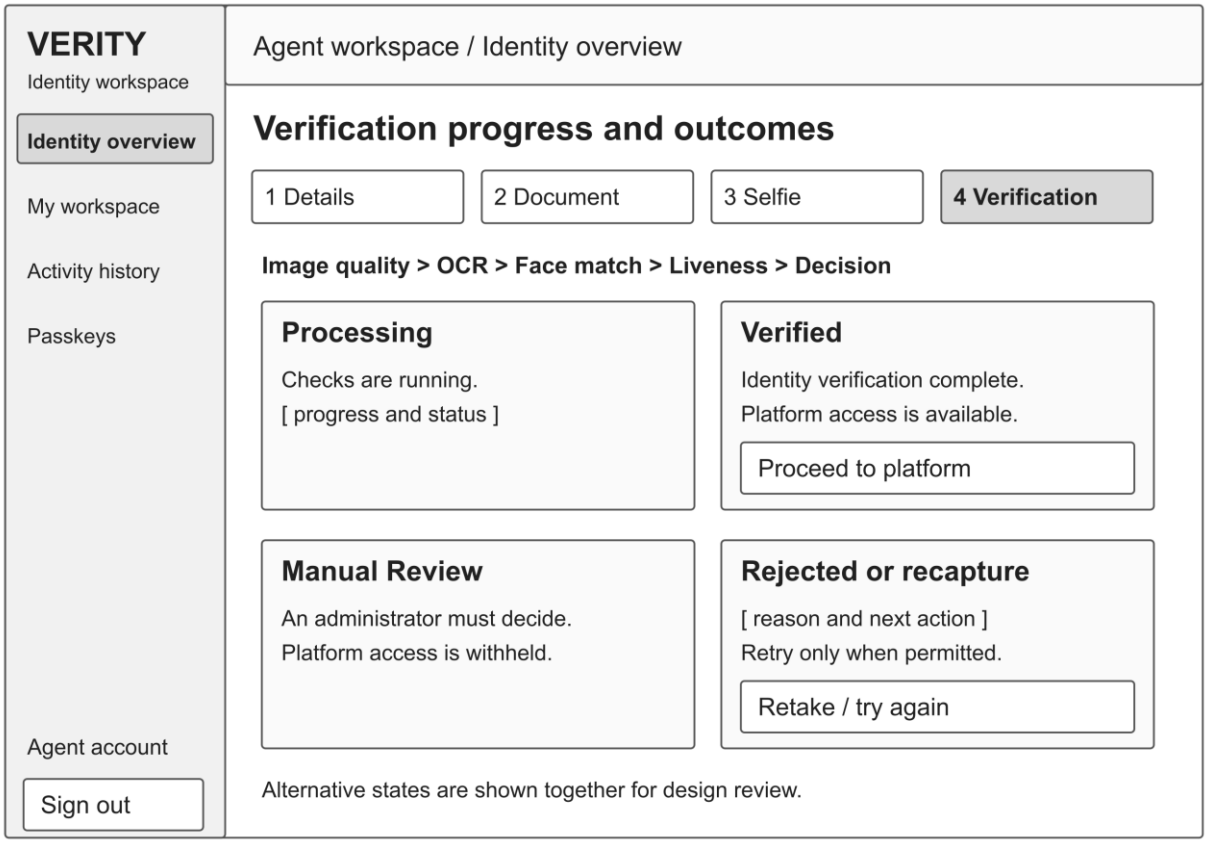



**Figure 4.11: Verification progress and decision states wireframe**

Only a verified outcome with an authorised access grant exposes the protected-workspace action. Review explains that an administrator must decide. Rejection and capture failure show the reason and a permitted next action; the interface must distinguish retaking poor evidence from an identity decision. Model decisions and threshold fallback must not be explained as the same rule. A retry is offered only while the backend permits it, and a failed integration does not show a success state.

#### 4.4.6.6 Administrator directory

The administrator directory provides an overview of agents, review workload and current model status. Role-specific navigation connects the directory with the review queue, recent verified cases, flagged attempts, audit records and model evaluation.

*Requirements addressed: FR-19, FR-21, FR-23 and FR-33; NFR-05; DR-06.*

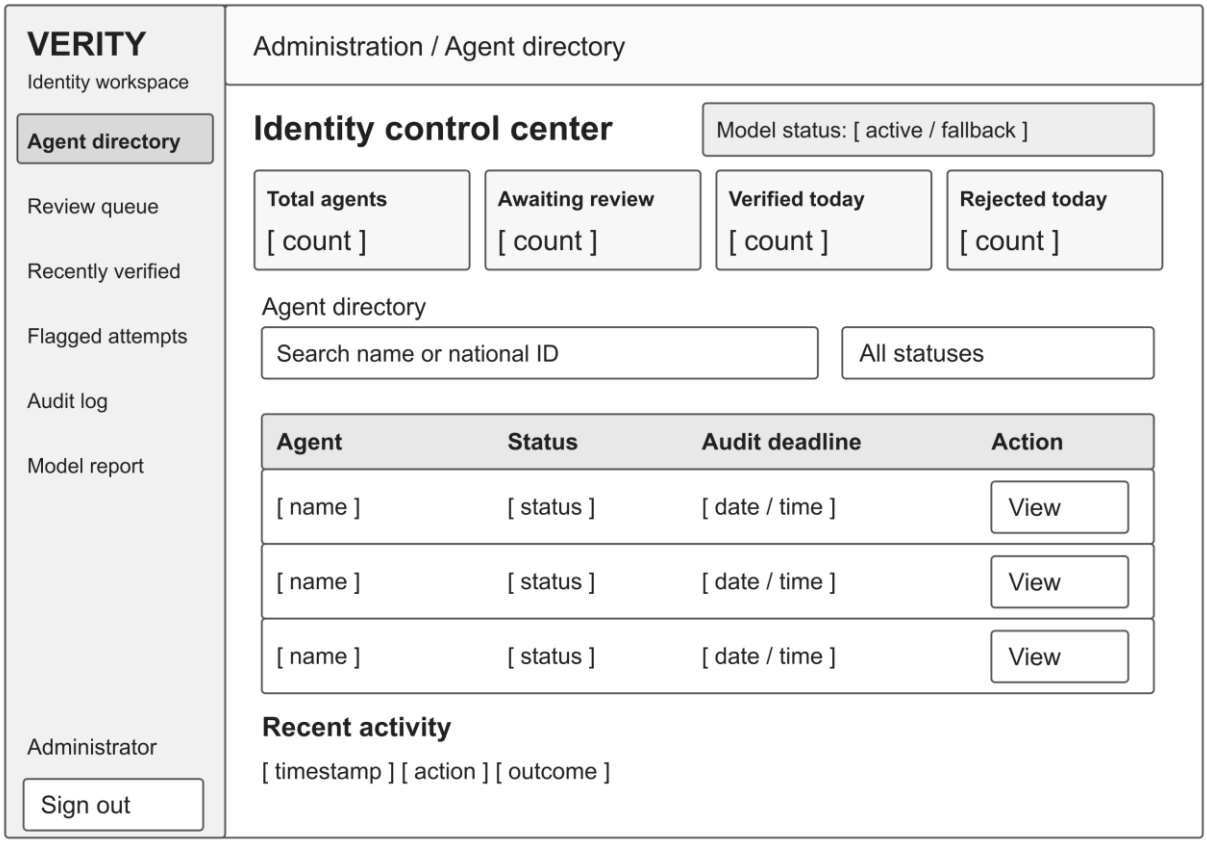



**Figure 4.12: Administrator directory wireframe**

The search and status controls narrow the displayed agent set. Each row provides status, an audit deadline when applicable and a route to case detail. Recently verified cases use the three-business-day image window, while the broader directory retains record history after images expire. Real-time review notifications invite the administrator to open a case; the persisted queue remains the reference if a notification is missed. Counts in the wireframe are placeholders, not study results.

#### 4.4.6.7 Administrator case review

The case-review wireframe groups identity images, registered and extracted fields, the six scores, decision provenance and available administrator actions. It abstracts the existing detail drawer into a readable design view.

*Requirements addressed: FR-19 to FR-21 and FR-24; NFR-13 and NFR-14; DR-03 to DR-07.*

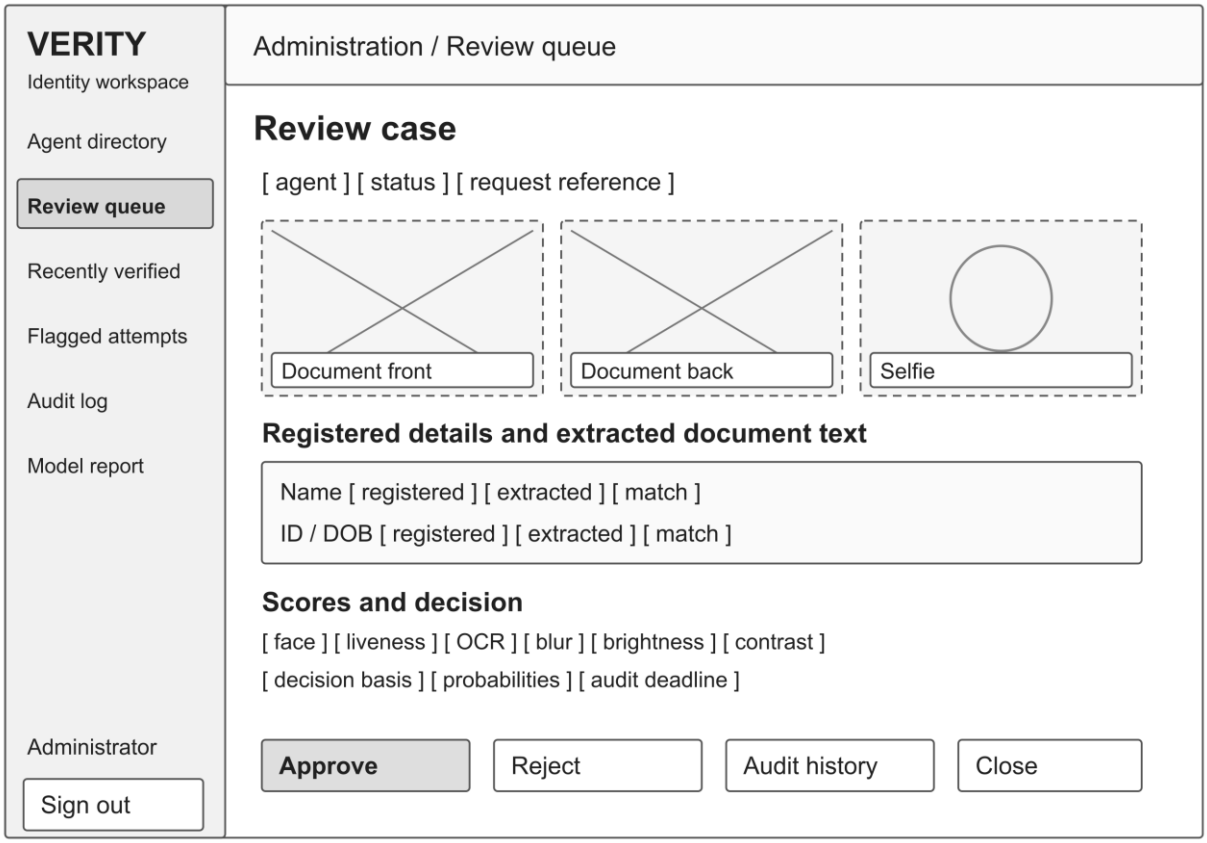



**Figure 4.13: Administrator case review wireframe**

The reviewer can compare evidence before deciding, inspect the audit history and close the case without making a change. Approve and Reject are authorised backend actions. A rejection dialog may record a reason; actor, time and outcome must be retained for either decision. An expired image is labelled unavailable while its metadata remains visible. The design does not imply that an administrator can recover a deleted image or that a model probability alone explains the underlying identity evidence.

#### 4.4.6.8 Audit history and export

The audit wireframe makes actions, outcomes and responsible actors available for administrator inspection. Pagination supports navigation through the event history, and a separate export dialog defines the requested subset.

*Requirements addressed: FR-22, FR-24 to FR-26; DR-05; NFR-05.*

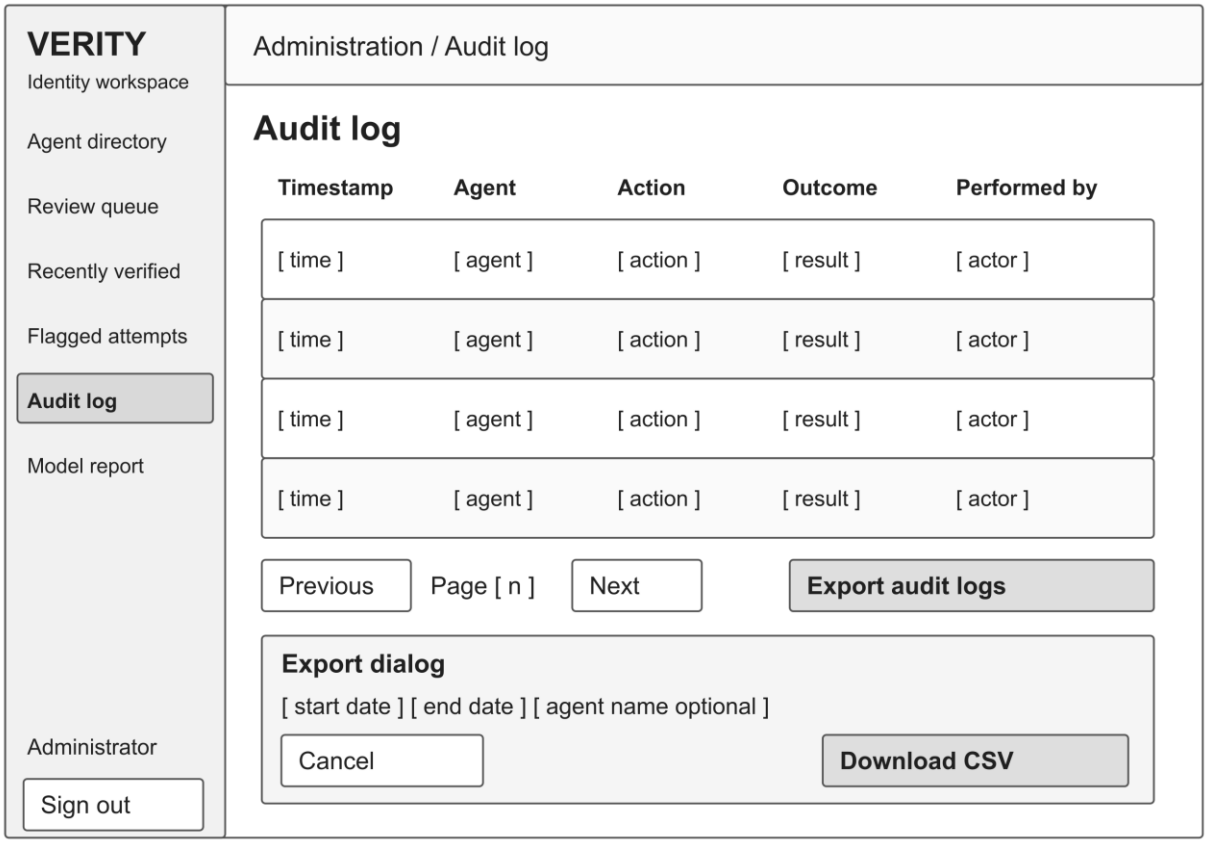



**Figure 4.14: Audit history and export wireframe**

Date and optional agent filters determine the CSV output. Export retains server-side role checks and exposes a readable failure if the request cannot complete. The log view contains no editing or deletion control for ordinary administrator use. Empty results must be distinguished from loading errors. The dialog is shown alongside the log for documentation and would appear as an overlay during use.

#### 4.4.6.9 Passkey management

The passkey wireframe provides account-level enrolment and credential management outside the evidence-capture flow. It identifies the registered device credentials without displaying secret key material or local device biometric templates.

*Requirements addressed: FR-28 to FR-30; IR-08; NFR-19.*

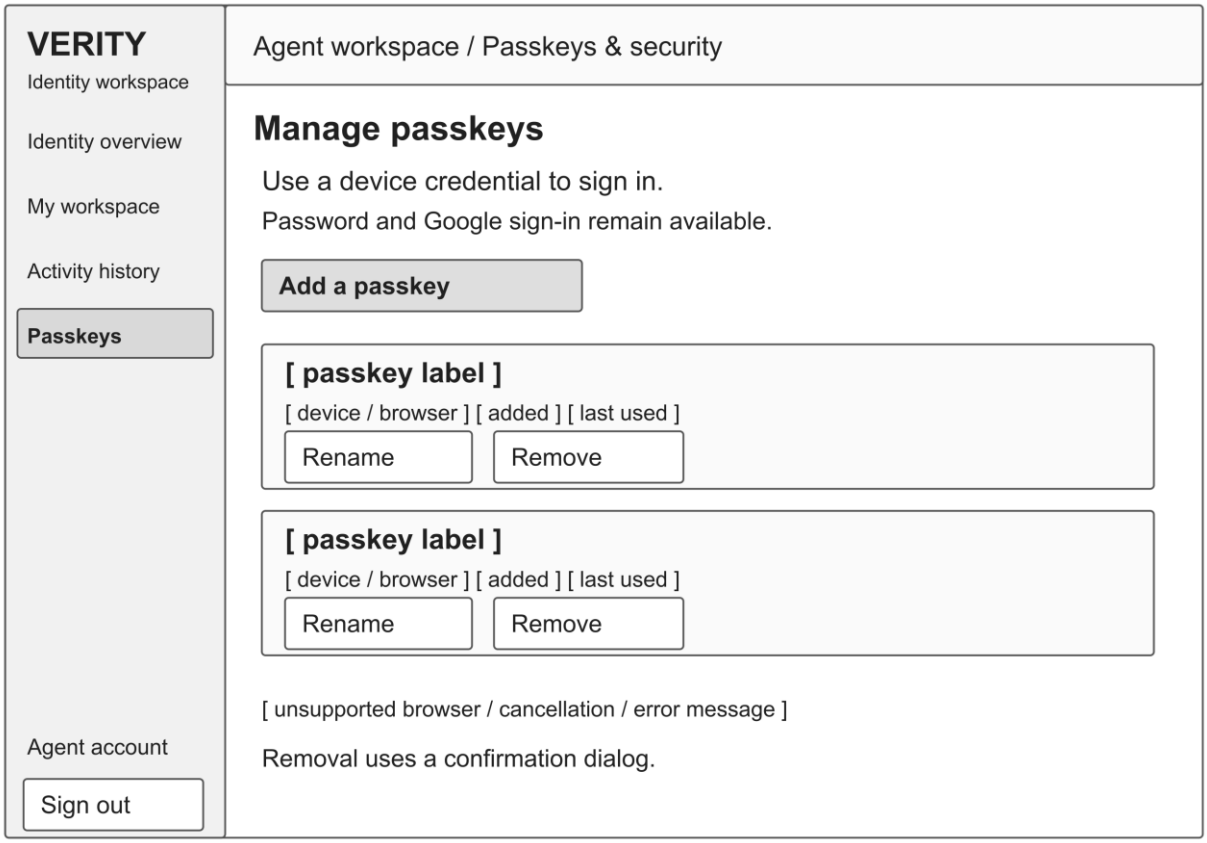



**Figure 4.15: Passkey management wireframe**

Each credential shows its label, device metadata and relevant dates, with Rename and Remove actions. Removal opens a confirmation dialog and remains subject to available recovery methods. Unsupported devices retain existing authentication routes. Loading, cancellation and failure messages are exposed in a status region. A successful registration does not bypass identity verification or grant access to the verified workspace.

#### 4.4.6.10 Model evaluation interface

The evaluation wireframe presents the saved model report to an administrator, including aggregate metrics, class-specific results and the confusion matrix. It links the report to its model and dataset context.

*Requirements addressed: FR-34; NFR-16 to NFR-18; IR-10.*

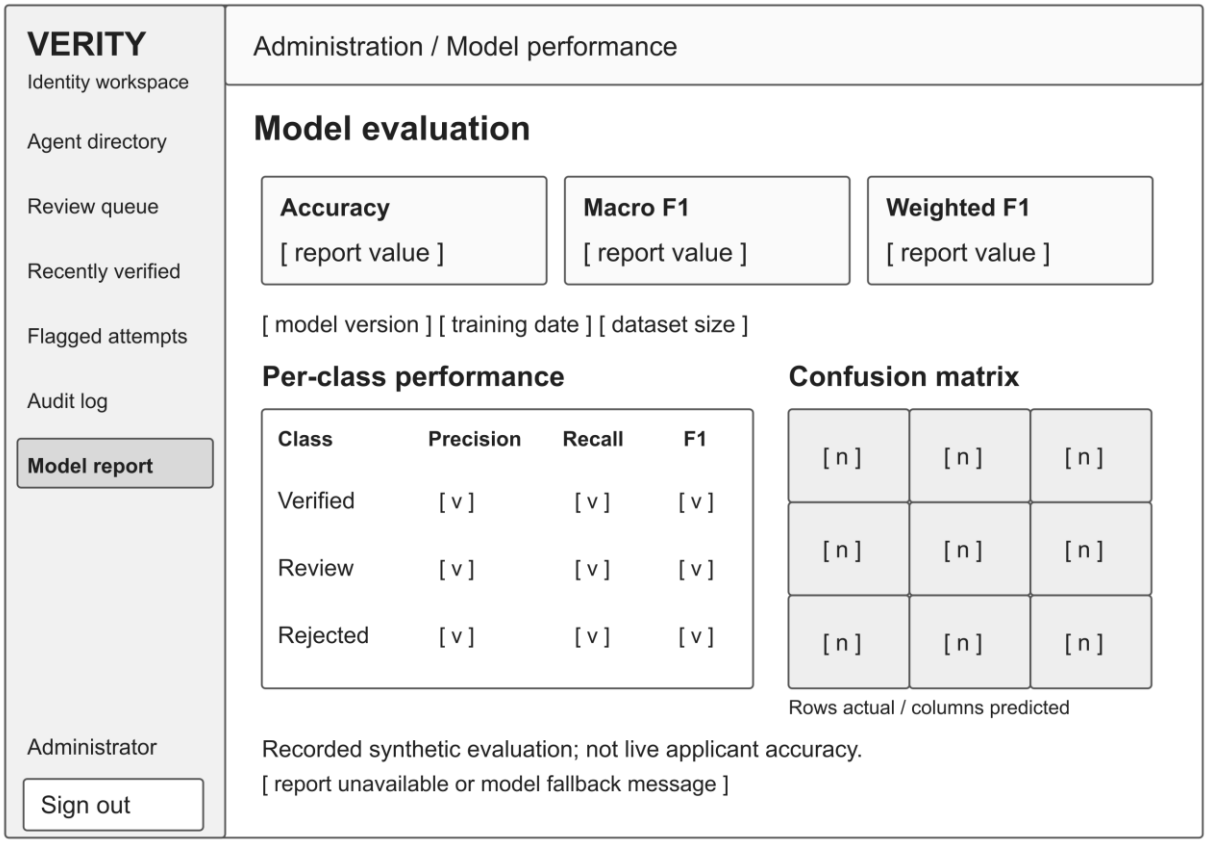



**Figure 4.16: Model evaluation interface wireframe**

The view distinguishes saved synthetic evaluation from current applicant outcomes and exposes a fallback or unavailable-report state. Cells contain placeholders rather than fabricated measurements. False-verification errors and weak Manual Review recall must remain visible in the accompanying evaluation discussion. The interface supports inspection; it does not initiate online training or establish that a hosted service is running the same model version as a local report.

### Interface states and responsive behaviour

Each asynchronous screen distinguishes loading, empty, success and failure states. The interface keeps the relevant error beside the affected task, avoids duplicate submission while a request is active and preserves a route to recovery. It must not expose protected content simply because a client button is visible; access is enforced by the backend. Text labels supplement colour, navigation exposes the current location and modal focus returns to its trigger on dismissal.

On a phone, the navigation menu precedes the task, form columns become a single sequence and image previews retain their aspect ratio. Wide administrator tables require controlled horizontal navigation or an equivalent readable presentation. Both light and dark modes require contrast and focus checks. Camera and human-check accessibility require device and assistive-technology evaluation; a schematic alone cannot establish inclusive operation.

**Table 4.8: Interface state coverage**

| View group | Important state or exception | Verification |
|---|---|---|
| Authentication | Unconfirmed email; failed credentials; cancelled passkey; recovery link | C4-T01, C4-T14 |
| Profile and capture | Invalid field; duplicate ID; denied camera; poor image; partial recapture | C4-T02 to T05 |
| Verification | Processing; verified; review; rejected; unavailable service | C4-T08, C4-T09, C4-T18 |
| Administrator | Empty queue; new review event; expired images; authorised intervention | C4-T11, C4-T12, C4-T22 |
| Audit and model report | No matches; failed export; missing report; active/fallback distinction | C4-T12, C4-T13, C4-T16 |

## 4.5 Requirements Traceability and Verification Plan

Traceability connects the problem and objectives to a requirement, a design view and a reproducible check. The objective references below follow the five objective themes used in the project evaluation: O1 investigates existing registration and verification practice; O2 identifies challenges and fraud risks; O3 examines existing KYC platforms; O4 develops the system; and O5 tests and validates effectiveness. O1 and O3 supply research context through the literature review, while O2, O4 and O5 directly drive the software and evaluation requirements.

Table 4.9 maps the complete requirement catalogue to the design and the verification scenarios in Table 4.10. The C4-T identifiers are Chapter 4 planning references. They do not renumber or claim to be the TC identifiers already used in Chapter 5. In the table, T02 after C4-T01 is shorthand for C4-T02, and the same prefix applies to the remaining scenario numbers.

**Table 4.9: Requirements traceability matrix**

| Requirement group | Objective and need | Design reference | Scenario |
|---|---|---|---|
| FR-01 to FR-06; FR-27 | O2, O4; A/O | 4.2.1; Figures 4.7-4.10 | C4-T01 to T04 |
| FR-07 to FR-12; FR-31 | O2, O4; A/O | 4.2.1; Figures 4.2, 4.5, 4.9-4.11 | C4-T05 to T07 |
| FR-13; FR-34 | O4, O5; B/R | 4.2.3; Figures 4.4, 4.16 | C4-T08, T16, T21 |
| FR-14 to FR-16; FR-32 | O2, O4; A/O | 4.4.4; Figure 4.11 | C4-T09, T15 |
| FR-17 and FR-18 | O2, O4; O | 4.2.1 guardrails | C4-T10 |
| FR-19 to FR-21; FR-33 | O4, O5; B | Figures 4.12 and 4.13 | C4-T11 |
| FR-22 to FR-26 | O2, O4; B/O | 4.4.5; Figure 4.14 | C4-T12, T13 |
| FR-28 to FR-30 | O4; A/O | 4.2.4; Figures 4.7 and 4.15 | C4-T14 |
| NFR-01 to NFR-03 | O5; A/O | 4.2.2; 4.2.4 | C4-T17, T18 |
| NFR-04 to NFR-08 | O2, O4; A/B/O | 4.2.4; 4.4.5 | C4-T01, T09, T19 |
| NFR-09 and NFR-10 | O2, O4; O | 4.2.4; 4.4.4 | C4-T08, T18 |
| NFR-11 and NFR-12; NFR-19 and NFR-20 | O4, O5; A/B | 4.4.6 all wireframes | C4-T05, T15 |
| NFR-13 and NFR-14 | O2, O4; B/O | 4.4.5; Figure 4.13 | C4-T07, T08, T11, T13 |
| NFR-15 | O4; O | 4.4.1; 4.4.3 | C4-T20 |
| NFR-16 to NFR-18 | O5; R/B | 4.2.3; Figure 4.16 | C4-T21 |
| DR-01 to DR-05 | O2, O4; A/B/O/R | Tables 4.5 and 4.7 | C4-T02, T03, T07, T08, T13 |
| DR-06 to DR-09 | O2, O4; A/O | 4.2.3; Figures 4.12 and 4.13 | C4-T15, T22 |

| Requirement group | Objective and need | Design reference | Scenario |
|---|---|---|---|
| DR-10 and DR-11 | O5; R | 4.2.3 experimental data design | C4-T23 |
| DR-12 | O4, O5; O | 4.2.3 recovery policy | C4-T24 |
| IR-01 and IR-09 | O4; A | 4.4.6; Figures 4.8-4.10 | C4-T04, T15 |
| IR-02 and IR-04 | O2, O4; O | 4.2.4 authentication boundaries | C4-T01, T09, T19 |
| IR-03 and IR-05 | O4, O5; O | 4.2.4; Figure 4.4 | C4-T18, T20 |
| IR-06 and IR-07 | O4; A/B/O | 4.2.4; Figures 4.12-4.14 | C4-T03, T12, T19 |
| IR-08 and IR-10 | O4, O5; A/B | Figures 4.15 and 4.16 | C4-T14, T16 |

**Table 4.10: Planned verification scenarios**

| Scenario | Focus | Required evidence |
|---|---|---|
| C4-T01 | Account and session | Valid and invalid login, confirmation, recovery, expired tokens and protected endpoints. |
| C4-T02 | Profile and guardrails | Valid profile, missing/invalid values, duplicate national ID and blocked email domains. |
| C4-T03 | Evidence storage | JPEG/PNG, 5 MiB boundary, document type/sides, private object ownership and upload failure. |
| C4-T04 | Live capture | Camera permission, framing, challenge completion, failure and browser/device compatibility. |
| C4-T05 | Quality and recapture | Clear/degraded evidence, downstream gate, explanatory feedback and targeted recapture. |
| C4-T06 | Evidence processing | OCR agreement, matching/non-matching faces and tested genuine/spoof presentations. |
| C4-T07 | Persistence and provenance | Linked scores/decisions/access records, partial attempts, key integrity and decision basis. |
| C4-T08 | Classifier contract | Six valid features, invalid/missing values, class-probability mapping and SVM fallback. |
| C4-T09 | Role and access | Verified/review/rejected access, unauthorised administrator calls and cross-agent identifiers. |
| C4-T10 | Repeated attempts | IP and fingerprint limits, configured 24-hour window and limit boundary. |
| C4-T11 | Administrator workflow | Directory/search, review/override, images, audit deadline and attempt history. |
| C4-T12 | Notification and export | Review-state events, reconnect, recipient access, CSV date/agent filtering and failure. |
| C4-T13 | Audit accountability | Required events, actor/context, early nullable references and forbidden client mutations. |

| Scenario | Focus | Required evidence |
|---|---|---|
| C4-T14 | Passkeys | Enrolment, signatures, challenge expiry/replay, ownership, rename, removal and recovery. |
| C4-T15 | Interface and accessibility | Wireframe tasks, state feedback, keyboard/focus, mobile layout, disclosure and assistance. |
| C4-T16 | Model report interface | Correct model/data labels, active/fallback state and unavailable report. |
| C4-T17 | Performance measurement | Declared hosts/network, cold/warm runs, whole pipeline and representative API timings. |
| C4-T18 | Failure and readiness | Service/model unavailability, timeout, invalid response, partial persistence and safe retries. |
| C4-T19 | Data protection | HTTPS, API credentials, client ownership policies, private objects and signed-link expiry. |
| C4-T20 | Service contracts | Independent image/score endpoints, defined inputs, outputs, errors and compatibility. |
| C4-T21 | Model evaluation | Held-out metrics, confusion matrix, false verification, calibration and stated limitations. |
| C4-T22 | Retention | Business-day expiry, repeat approvals, cleanup retries, metadata and non-verified policy scope. |
| C4-T23 | Experiment reproducibility | Configuration/seed/source records, duplicates, split isolation and fitted preprocessing. |
| C4-T24 | Recovery | Controlled backup, isolated restoration and record/model integrity without premature data deletion. |

## Connection to Chapter 5 evidence

The Chapter 5 classifier-contract tests TC01 to TC04 and TC07 to TC09 provide evidence relevant to C4-T07, C4-T08 and part of C4-T18. Its experiment checks TC05 and TC06 relate to C4-T23. The recorded passkey and retention suites support parts of C4-T14 and C4-T22. Passing an isolated suite does not demonstrate the complete hosted user workflow, a real-device ceremony or live storage cleanup.

System-level verification requires dated records from the configured environment, the software/model revision, input conditions, expected and actual results, and the requirement identifier. Latency, demographic fairness, presentation-attack coverage, accessibility, informed disclosure and restoration must remain unverified until their corresponding evidence exists. If a requirement changes, its affected interface, data rules, diagrams and tests must be reviewed together. Changes to approved project scope are discussed with the supervisor.

## 4.6 Chapter Summary

The design connects account registration and controlled evidence capture with six-score processing, an explicit decision basis and role-dependent access. Administrator review, audit history, bounded image retention and optional passkeys support the main workflow. The requirements catalogue, logical entity description, interface wireframes and traceability plan establish what the implementation and evaluation must demonstrate. Chapter 5 reports the implemented environment, actual tests, classifier results and remaining evidence limitations.

## References

ISO/IEC/IEEE. (2018). ISO/IEC/IEEE 29148:2018 Systems and software engineering - Life cycle processes - Requirements engineering.

https://www.iso.org/standard/72089.html

ISO/IEC. (2023). ISO/IEC 25010:2023 Systems and software engineering - Systems and software Quality Requirements and Evaluation - Product quality model.

https://www.iso.org/standard/78176.html

World Wide Web Consortium. (2023). Web Content Accessibility Guidelines 2.2 and supporting quick reference.

https://www.w3.org/WAI/WCAG22/quickref/

Project design evidence: the maintained application source, database migrations and version 3 experiment records, together with Chapter 5 System Implementation Testing and Evaluation, hosted deployment revision dated 27 October 2026. Model-development details are recorded in the project's machine learning README, experiment protocol, split manifests and saved reports. The wireframe source-to-screen mapping is retained with the editable figure files.